Photoacoustic Image Reconstruction Using ResNet and Swin Transformer Models

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1 Method Overview

We study the problem of reconstructing optical absorption maps from dual-wavelength photoacoustic pressure signals acquired at 755 nm and 808 nm. The task is formulated as a supervised image-to-image regression problem in which the model learns a mapping from the pressure domain to the absorption domain.

1.1 Baseline Model

The baseline architecture is a ResNet18 encoder-based U-Net. A convolutional backbone pretrained on ImageNet is used for hierarchical feature extraction, and a U-Net-style decoder reconstructs the absorption maps through progressive upsampling with skip connections.

- **Encoder:** ResNet18
- **Decoder:** U-Net style convolutional decoder
- **Loss:** L_1 loss
- **Input:** 2-channel pressure maps (755 nm and 808 nm)
- **Output:** 2-channel absorption maps

1.2 Proposed Model: SwinV2 U-Net

The proposed model replaces the CNN encoder with a Swin Transformer V2 Tiny backbone implemented with `timm`, while keeping the decoder unchanged for a fair architectural comparison. The encoder is used in `features_only=True` mode so that multi-scale feature maps can be passed directly to the same U-Net-style decoder used in the baseline.

- **Encoder:** SwinV2 Tiny Transformer

- **Decoder:** Same U-Net style decoder as the baseline
- **Loss:** L_1 loss
- **Input/output format:** Unchanged from the baseline

The motivation is to assess whether global attention improves reconstruction quality for spatially complex biological structures that may benefit from longer-range contextual reasoning.

2 Quantitative Results

2.1 Test Set Performance

Table 1 summarizes the test-set performance of the baseline and proposed models.

Table 1: Test set performance comparison. Lower is better for MAE and MSE; higher is better for PSNR and SSIM.

Model	**MAE** $\downarrow$	**MSE** $\downarrow$	**PSNR** $\uparrow$	**SSIM** $\uparrow$
ResNet18 (Baseline)	0.02075	0.00167	29.35	0.8923
SwinV2 (Proposed)	0.02039	0.00145	29.60	0.8604

2.2 Observations

The SwinV2 model slightly improves mean absolute error, reducing MAE from 0.02075 to 0.02039. It also improves PSNR from 29.35 dB to 29.60 dB, indicating better overall signal fidelity. However, the ResNet18 baseline achieves higher SSIM, suggesting that it preserves local structural characteristics more effectively.

Overall, the results indicate a trade-off between global feature modeling and local spatial consistency: SwinV2 improves pixel-level accuracy and signal fidelity, whereas ResNet18 better preserves structural similarity.

2.3 Metric Distribution

To complement the aggregate test-set metrics, Figure 1 shows the distribution of reconstruction performance across samples for both models. These boxplots help reveal the spread, central tendency, and potential outliers that are not visible from mean values alone.

The boxplots provide a distribution-level comparison between the two models. In particular, they help assess whether the improvement in average MAE and PSNR for SwinV2 is consistent across samples or driven by a subset of easier cases, while also highlighting the relative variability in SSIM.

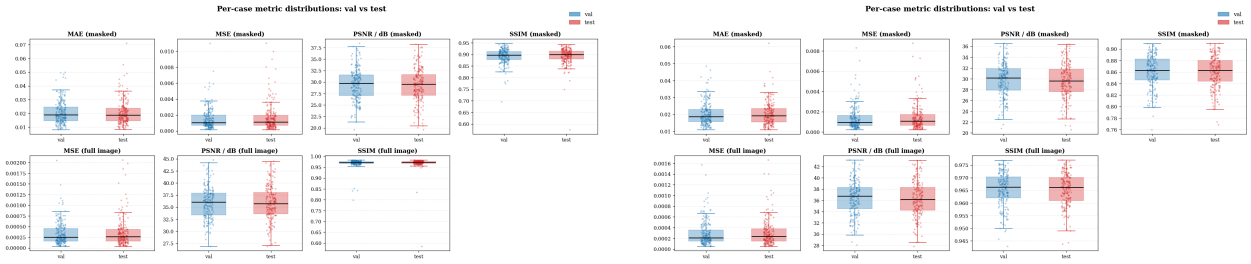

Figure 1: Metric distribution boxplots for the baseline ResNet18 model (left) and the proposed SwinV2 model (right).

2.4 Validation Set Performance

Validation metrics can be inserted here in the same format as Table 1 if they are available in `eval_results`.

3 Qualitative Results

Qualitative evaluation is performed by visualizing reconstructions at the 0th, 25th, 50th, 75th, and 100th percentiles of MAE. For each selected sample, the visualization includes:

- input pressure maps at 755 nm and 808 nm,
- ground-truth absorption maps,
- predicted absorption maps, and
- absolute error maps.

The qualitative trends are consistent with the quantitative metrics. Low-error samples show near-perfect structural alignment between prediction and target. High-error samples tend to occur in regions with weak or noisy pressure signals, where the inverse mapping becomes less stable. Visually, SwinV2 tends to produce smoother reconstructions, while ResNet18 often preserves sharper local details.

If available, the percentile visualization exported by `evaluate.py` can be inserted here as a figure.

4 Comparison to Baseline

Table 2 reports the absolute change of the proposed model relative to the baseline.

The improvement in MAE and PSNR suggests that the transformer-based encoder produces more accurate reconstructions on average. In contrast, the reduction in SSIM indicates a modest loss in structural fidelity. This pattern suggests that SwinV2 captures broader spatial dependencies

Table 2: Absolute metric differences for SwinV2 relative to ResNet18.

Metric	Change
MAE	$\downarrow$ 0.00036
PSNR	$\uparrow$ 0.25 dB
SSIM	$\downarrow$ 0.0319

effectively, but may smooth out fine-grained anatomical or textural structures that CNNs preserve better.

5 Discussion

5.1 Why SwinV2 Improves MAE and PSNR

The SwinV2 encoder can model long-range dependencies across the image, which helps integrate contextual information over larger spatial extents. For photoacoustic reconstruction, this broader receptive field may improve global consistency in the recovered absorption maps, especially in regions where local evidence is weak or ambiguous.

5.2 Why SSIM Decreases

CNN-based encoders such as ResNet18 encode strong local spatial priors through convolution and weight sharing. These inductive biases often help preserve edges, local textures, and fine structural detail. By comparison, the global attention mechanism in SwinV2 may favor smooth, coherent predictions at the expense of subtle local variations, which can reduce SSIM even when pixel-wise error improves.

6 Failure Cases

The worst-performing samples, corresponding to the highest-MAE percentile, typically exhibit one or more of the following characteristics:

- weak or noisy input pressure signals,
- low-contrast regions in the absorption targets, and
- ambiguity in the mapping between wavelength channels and absorption patterns.

These cases highlight the difficulty of reconstructing low-SNR regions and suggest that performance remains constrained when the measured signal contains insufficient information.

7 Conclusion

We compared a ResNet18-based U-Net baseline with a SwinV2-based encoder-decoder model for photoacoustic image reconstruction from dual-wavelength pressure inputs. The SwinV2 model improves MAE and PSNR, demonstrating stronger average reconstruction accuracy and signal fidelity, but it reduces SSIM relative to the CNN baseline.

In summary, SwinV2 is a competitive alternative to CNN-based encoders for this inverse imaging task. Its performance suggests that transformer-based global context modeling is beneficial, although preserving fine local structure remains an important challenge for future work.